Evidence-Supported ASO Catalog

An attributed orbital catalog for Space Data Network

Anthony "TJ" Koury III and Dr. Moriba Jah

Koury: Edgesource, Space Data Network. Jah: The University of Texas at Austin; GaiaVerse Ltd.

Technical whitepaper 2.0 | 11 October 2026

Numerical evidence cutoff: 11 October 2026

1 Summary

Space Data Network (SDN) builds a catalog of anthropogenic space objects (ASOs) from public and login-only sources. It keeps every provider's product as an attributed record and, object by object, publishes the solution with the best measured accuracy for the intended use, with its uncertainty and lineage. Accuracy is measured against independent precise orbits, under plans fixed before the test data were read R1.

Object class	Catalog product	Measured (median)
GPS satellites	Analysis-centre orbit, propagated	3.5 cm at issue, 30 m a day later
Satellites with laser ranging	High-precision orbit propagation (HPOP) fitted to the ranges	7 m at the fit, 42 m after 7 days
Satellites with operator ephemerides	The operator's ephemeris	Public element sets (OMMs) lie 0.6–5 km from it
Objects known only from public OMMs	SGP4 on the OMM	0.3–1.6 km at issue

HPOP's force models agree with Orekit within 1.6 cm over a day, and from precise states they leave a median 4.8 m against laser-ranging orbits after a day. Started from an OMM's own epoch state, HPOP is never better than SGP4: the start state, not the force model, limits it. None of the density corrections tested improves 1–3-day orbit prediction in its pre-registered test, and covariances proved realistic on held-out data in few regimes and horizons.

Three results go beyond published work and are marked [NEW] where they enter: GPS element-set correction fails on an unseen orbital plane (section 5.3); a density correction of the kind the Space Force's drag model uses, solved from Planet's public states, is scored on satellites that did not calibrate it (section 5.4); consecutive OMM errors are strongly correlated, and four published covariance constructions are overconfident (section 6). Pre-registered hypotheses that failed are reported as not supported.

2 Evidence model and records

An orbit solution is an inference with five contributors: the object's physics, the dynamics model, the observations, the measurement model and the inference method. A provider's ephemeris or element set already embeds the provider's models and inference, so the catalog keeps it as an attributed provider claim, not as an observation. Agreement between products counts as independent confirmation only when their lineage is independent; unknown lineage is recorded as unknown.

Orbit data sit at four processing levels, by analogy with Earth-science data levels. A higher level is more processed, not better: it inherits every flaw below it, and re-epoching adds no evidence. Every source record declares its level.

Level	Content	SDN inputs
0	Raw sensor data	Detector counts, signal samples, images
1	Calibrated, time-tagged measurements	Laser normal points; radio Doppler; optical angles
2	Estimated states and ephemerides	Operator ephemerides; GNSS and laser-ranging precise orbits; Vimpel elements; ILRS predictions
3	Theory-specific mean elements fitted to level 1 or 2	OMMs from Space-Track and SpaceMapper; CelesTrak's supplemental GP; SDN's SGP4 companions

A source record keeps the provider, native identifier, edition, SHA-256 of the bytes, product type, epoch, time scale and frame; observation time, solution epoch, issue time and retrieval time stay distinct. A solution record specifies:

Field	Content
State and validity	State or ephemeris, epoch, applicable interval
Evidence and lineage	Source records, their levels and dependencies, editions
Dynamics	Force model, environment inputs, parameters estimated, considered or assumed
Measurement model	Sensor geometry, timing, calibration, biases; unknown when a provider does not state it
Inference	Estimator or selection rule, weights, priors, software version
Uncertainty	Covariance or none, its source and its calibration label

Hashes establish byte identity and signatures establish attribution; physical validity rests on the measurements of sections 5 to 7. Identical inputs, module binaries and policy reproduce the same decisions.

3 Sources

The inventory R2 lists 97 public and login-only sources in eight groups, with access, terms, latency and use. Results draw on every source an experiment read; files are copied only where their terms allow, and the credits close this paper.

Group	Sources	Role
A. Precise orbits	IGS and ESA GNSS orbits; ILRS orbits and normal points; Sentinel-1, Swarm, GRACE-FO and CryoSat-2 precise orbits	Truth; the GNSS and laser-ranged products
B. Operator ephemerides	SpaceX Starlink (with covariance), Planet, NASA ISS, CMSE Tiangong, Intelsat; ILRS predictions	Operator products; drag and maneuver evidence
C. Catalogs and observations	Space-Track GP history (OMM) and SATCAT; JSC Vimpel; SpaceMapper AOE; SatNOGS Doppler	Element sets; other providers' catalogs; public observations
D. Density	Swarm and GRACE-FO densities; CHAMP and GRACE-A densities with the High Accuracy Satellite Drag Model (HASDM) R3	Density truth; the HASDM comparison
E. Space weather	Space Environment Technologies' (SET) JB2008 indices; GFZ Kp, ap and F10.7; NOAA SWPC forecasts	Model drivers, as issued
F. Earth orientation, gravity, ephemerides	IERS EOP; EGM2008, EGM96; FES2004; DE440	Frames and dynamics

Group	Sources	Role
G. Physical properties	GCAT; IGS satellite metadata	Priors for drag and radiation parameters
H. The High Accuracy Catalog (HAC) and HASDM	Special Perturbations (SP) vectors; real-time HASDM	The comparison target (section 11)

SpaceMapper's AOE catalog (channel KY) holds 19,197 element sets that the provider attributes to its own sensor network R4. AOE rewrites them every 3 hours; SDN takes a snapshot daily from the public web pages. SpaceMapper states no license, so the sets are cited, not copied. Doppler read from SatNOGS waterfall images made the ISS orbit 2.7–10.7 times worse than SGP4 E6.

4 Normalization and model dispatch

Vimpel osculating elements

Vimpel's HEO product gives osculating elements in J2000 R5. The standard perifocal rotation turns them into a J2000 state at the same epoch R6, with $\mu = 398{,}600.4418$ km^3/s^2, under the identifier `vimpel:<id>` and no invented NORAD number R7. Velocity differenced from 600 s position samples departs from the analytic velocity by up to 1.1 km/s, so it serves only as a diagnostic.

The handoff contract

An OMM holds SGP4 mean elements, not osculating ones. The catalog evaluates SGP4 at zero elapsed time, transforms position and velocity from TEME to GCRF at the same instant, and starts HPOP from that state R8. SGP4's drag term B* becomes the ballistic parameter

$$B = \frac{C_D A}{m} = \frac{2B^*}{\rho_0}$$

with ρ_0 from Spacetrack Report No. 3 R9; 0.01 m^2/kg stands in where $B^* \leq 0$, and Cr $\cdot$ A/m is 0.02 m^2/kg R10. The handoff is valid; whether it helps is measured in section 5.3.

Levels of measurement of OMM elements

OMM fields mix nominal (catalog number), ordinal (element-set number), interval (epoch), ratio (mean motion) and circular (node, perigee, mean anomaly) scales, so the pipeline computes on SGP4-evaluated states, never on raw fields. The legacy two-line format quantizes angles to 0.0001°, about 12 m along track near 6,900 km: a resolution, not an accuracy.

Dispatch, frames and time

Each record is routed by its declared theory: mean elements to SGP4, osculating elements to the two-body conversion, states and ephemerides to frame transforms. Osculating elements offered to SGP4 are refused as a model mismatch. Frames follow the IERS Conventions (2010): IAU 2006/2000A, CIO-based, through ERFA, with IERS Earth orientation R11. TEME, EME2000, GCRF and ITRF stay distinct; epochs are UTC and integration runs on TT.

5 Products by object class

Each row is one method: its originators, our measured result (median 3D position error unless stated) and the experiment. Every experiment froze its plan before reading its test window R1.

5.1 GNSS and laser-ranged satellites

Method	Prior art	Measured	Evidence
Newest ultra-rapid analysis-centre orbit at issue, propagated numerically beyond	IGS products R12	GPS, 32 satellites: 3.5 cm at issue, 336 m at 7 days; the latest public OMM 1.5–1.7 km	E3
HPOP fitted to ILRS laser normal points	Laser-ranging orbit determination R13	Starlette, Stella, LARETS, WESTPAC, LARES: 7 m at the fit and 42 m at 7 days, against SGP4's 472 m and 685 m. One pass beats the OMM for a day; four passes beat it at every horizon to 7 days	E6

5.2 Operator- and provider-ephemeris objects

Method	Prior art	Measured	Evidence
Operator ephemeris, within its span	Operator orbit determination	GNSS analysis centres' predictions (IGS for GPS, ESA for GLONASS): 0.1 m from final orbits at 0–12 h	E4
SGP4 on the latest public OMM, against operator ephemerides	Dyreby et al. 2025 R14; Jankovic 2026 R15	At the file's start: Planet 0.59 km, ISS 0.82 km, GPS 1.23 km, Starlink 2.87 km, Intelsat 4.79 km; Starlink 13.8 km at 24 h	E4
Vimpel's state refined to Vimpel's ephemeris by one six-state correction	Weighted least squares R16	Withheld RMS 6.04–10.71 km before, 52.8–69.9 m after, on three 4-hour arcs: consistency with one provider, not accuracy	R7

5.3 Objects with public OMMs only

Method	Prior art	Measured	Evidence
SGP4 on the latest public OMM	Hoots and Roehrich 1980 R9; Vallado et al. 2006 R8	At issue: 0.31 km (laser-ranged LEO) to 1.57 km (GPS); by horizon in section 11	E3
Along-track correction at epoch (GPS)	Ly et al. 2020: 65 % R17; Hallgarten La Casta and Amato 2025 R18	RMS at epoch 2.04 → 0.79 km (61 % [53, 67]); the single-model form 8 %; worse than SGP4 by 7 days (2.55 against 1.88 km); its covariance fails the coverage gate. **[NEW]** On an orbital plane outside the fit: 6 % [2, 10]	E1
HPOP from the OMM's epoch state	Vallado et al. 2006 R8; Jankovic 2026 R15	Never better than SGP4 in four regimes (GPS, LEO with precise orbits, laser-ranged LEO and MEO); worse in all four by 72 h; at 24 h, 1.7 (GPS) to 11 (laser-ranged LEO) times SGP4's error	E7
HPOP fitted to an arc of OMM epoch states	Levit and Marshall 2011 R19	Better than SGP4 only at short horizons: 0.32–0.51 of its error to 12 h (laser-ranged LEO), 0.36–0.53 to 48 h on E1-corrected GPS sets; undecided beyond	E7

The start state sets the limit. Over 6 days, HPOP leaves 0.2 km along track from a GPS satellite's IGS state and 8.4 km from its OMM epoch state (medians of 508 arcs) R10. Laser ranges fix the start state, and the prediction then stays 20 to 60 times better than SGP4 to 7 days E6.

5.4 Drag regime

Density is scored by σ, the spread of ln(observed/model density) on satellites outside the calibration; prediction by the 3-day position error relative to JB2008's (below 1 is better).

Method	Prior art	Measured	Evidence
JB2008 with SET's indices	Bowman et al. 2008 R20	σ 0.155 (Swarm, GRACE-FO, 2026); 0.148 on CHAMP and GRACE-A in 2003 and 2008, where HASDM scores 0.078	E5
Temperature correction from accelerometer and precise-orbit densities, a stand-in for HASDM's calibration	Storz et al. 2005 R21; Doornbos 2012 R22	σ 0.056, 64 % [57, 70] lower; 0.112 on HASDM's windows; forecast gain 9 % for one day; 3-day ratio 1.04 [0.96, 1.10]; inputs arrive 49–92 days late (E8)	E5
Temperature correction from the OMM decay of 48 catalogued objects	Picone et al. 2005 R23; Gondelach and Linares 2020 R24; Gershman 2026 R25	σ 0.142, 8.4 % [−3.9, 18.3] lower (25 % required: not supported); 24.7 % lower in storms; as a forecast 16.8 % worse than JB2008; 3-day ratio 1.02; 0.270 on HASDM's windows	E8

[NEW] An open, DCA-form correction from public operator states E13. HASDM's Dynamic Calibration of the Atmosphere (DCA) corrects exospheric temperature every 3 hours from the drag of about 75–100 tracked calibration satellites, each with a constrained ballistic-coefficient correction R21. E13 solves that form from public data alone: the common mode s of the daily drag of 36 Planet Doves, read through orbital energy, corrects JB2008's temperature, with a ballistic coefficient held constant per satellite over a block and a forecast with a 48-hour time constant. Published work derives density from operator ephemerides and scores density alone R26, R27, R28, or assimilates commercial GPS states R29. E13 scores the correction on satellites that did not calibrate it, over ten months of Planet states against Swarm and GRACE-FO, with the storm of January 2026 in a held-out block. With x′ the logarithmic residual of observed against JB2008 density in a 12-hour window,

$$R_1 = 1 - \frac{\mathrm{sd}(x' - s)}{\mathrm{sd}(x')}$$

Pre-registered test (threshold)	Test A, Jan–Mar 2026, Kp 8.7 storm	Test B, Apr–Jul 2026
H1: Swarm and GRACE-FO density spread explained, R_1 (≥ 0.25)	0.247 [0.137, 0.347]: not supported	0.401 [0.283, 0.501]: supported
H3: held-out Doves' drag explained (≥ 0.2)	0.378 [0.223, 0.477]: supported	0.254 [0.135, 0.359]: supported
H4: Swarm position error ratio at 1 and 3 days (≤ 0.9)	1.02, 0.92: not supported	1.01, 1.00: not supported
H5: held-out Doves, the same	0.98, 0.97: not supported	1.00, 1.01: not supported
H6: control, correction shifted 7 days ($R_1 < 0.05$)	−0.149: passes	−0.082: passes

The correction does not follow storms. Exploratory, on one NRLMSISE-00 reference for every split: the fleet explains 0.47–0.48 of NRLMSISE-00's larger spread, leaving residuals of 0.075–0.095 against 0.069–0.086 on JB2008.

5.5 Maneuver detection

The detector compares the trajectories of consecutive element sets at one instant, whatever propagator answers its queries: a crossing test and a robust likelihood-ratio test for steps in orbit-averaged energy and orbit plane, merged by default R10. It builds on element-set time series R30 and the generalized likelihood ratio for jumps R31. Thresholds came from the ISS in 2020–2024; every other window was read once. A planned ISS burn counts as truth only when NASA's next trajectory shows it executed; one 2021 plan was not.

Held-out class, 2025–2026	Truth	Found	False alarms
ISS	25 burns	22; the misses are 0.1 m/s settling burns minutes before reboosts; median time error 10 min, Δv ratio 1.00	0.57 per year
Metop-SG A1	24 EUMETSAT maneuvers	14; 12 of 16 at 0.1 m/s or more	0
GPS, 4 satellites	6 maneuvers in precise orbits	6	3.8 per object-year
GEO (Intelsat)	11 post-maneuver files	4	24 events per object-year; the truth lists one maneuver per satellite

6 Uncertainty

Covariance propagates through HPOP's state-transition matrix Φ with a declared white-acceleration process noise Q R10:

$$P(t) = \Phi(t, t_0)\, P_0\, \Phi(t, t_0)^{\mathsf{T}} + Q(t)$$

Φ supplies neither P_0 nor Q; they come from a fit, a provider or an empirical model. Realism is scored against independent precise orbits by $d^2 = \mathbf{e}^{\mathsf{T}} P^{-1} \mathbf{e}$ for a position error e, against χ^2 with 3 degrees of freedom: a covariance is realistic when the mean of $d^2/3$ lies in [0.8, 1.25] and its 95 % coverage in [0.93, 0.97], the rule of E14 after Zaidi and Hejduk R32.

Covariance	Prior art	Held-out result	Evidence
Formal covariance of an HPOP fit to laser ranges	Weighted least squares R16	Too small: mean $d^2/3$ 16–21, coverage 0.14–0.18	E6
HPOP fit to OMM history alone	Levit and Marshall 2011 R19	Realistic for laser-ranged LEO (0.68–0.83, coverage 0.96–0.98); too small for the ISS within a day (4.8–7.6)	E6
Batch fit to 3 days of OMM history (GPS)	Osweiler 2006 R33; Geul et al. 2017 R34	Mean $d^2/3$ 1.23 and 1.05 at 0 and 1 day, 81 at 3 days; consistency failed at every horizon	E2

Covariance	Prior art	Held-out result	Evidence
[NEW] Errors of consecutive OMMs, against precise orbits	Autocorrelation shown by Osweiler R33	Along-track correlation 0.61–1.00 in four regimes. Four published constructions R33, R35, R34, R36 are overconfident: mean $d^2/3$ from 13 to over 10^7. A covariance that models the correlation scores better at 1 day in three regimes but is consistent only for laser-ranged satellites at some horizons	E2b
SGP4 with the E2b error model, calibrated by regime and age	Zaidi and Hejduk 2016 R32	Realistic in 2 of 35 regime-horizon cells (11 on the selection window)	E14
HPOP from the OMM epoch state, with P_0 and Q per regime	SP covariance as the Astrodynamics Support Workstation (ASW) forms it R37	5 of 35 (6 on the selection window): 500–800 km at epoch, above 800 km to 2 days	E14
Density-forecast consider term, in the form of the Dynamic Consider Parameter	Hejduk 2019 R38	No measurable effect in storms; density-error coverage 0.83–0.86 against 0.95	E14

On the held-out window, maneuvers, the storm of 4 July 2026 and GNSS repositioning put heavy tails on the errors: mean $d^2/3$ rises while medians and coverage stay near nominal in LEO, so realism found on the selection window did not carry over. Operator covariance enters as supplied, labeled with its source.

7 Dynamics and verification

HPOP's full force model, as the catalog and the conjunction screen run it R10, beside the SP options a Vector Covariance Message (VCM) names R39:

Model	HPOP	SP, as a VCM names it	Equivalence
Geopotential	EGM2008 to degree and order 70; EGM96 and tesseral truncation (36Z,24T) selectable	EGM-96 with zonal and tesseral degrees	EGM96 cases within 1 cm of Orekit
Atmosphere	NRLMSISE-00 (Orekit's to 1e-8); JB2008 (Orekit's to 2e-14); Jacchia-Roberts (GMAT's R40, bit for bit)	JBH09, which is JB2008 with HASDM's calibration R41; JAC70; MSIS90	JB2008 matches Orekit's; HASDM's calibration is not public (section 5.4); Jacchia-Roberts and NRLMSISE-00 are declared substitutes for JAC70 and MSIS90
Third bodies	Sun, Moon, Venus, Mars and Jupiter from DE440s	Sun and Moon	Sun and Moon within 1 cm of Orekit
Radiation	Cannonball with a conical shadow; Earth albedo and infrared	Solar radiation pressure, AGOM	Within 1 cm of Orekit
Tides, relativity	IERS 2010 solid tides and relativity; FES2004 ocean tides to 30×30	Solid Earth tides	Within 1 cm of Orekit
Drag parameter, thrust	$Cd \cdot A/m$ and its rate; in-track acceleration	B, BDOT; in-track thrust	Within 1.6 cm of Orekit

Model	HPOP	SP, as a VCM names it	Equivalence
Integrator	RK78 at 1e-12, steps of at most 300 s; its own error after 3 days 0.1–1.2 m	Eighth-order Gauss–Jackson on Cowell's equations R42, R43	Declared substitute, converged
Earth orientation	IAU 2006/2000A with daily IERS rows	IAU 1980 nutation, truncated	Declared substitute

Agreement with Orekit

HPOP's Orekit 13.1 suite R44 holds 72 cases on five orbits (400 km, 700 km sun-synchronous, GPS, geostationary, Molniya), each 24 hours, compared hourly on identical states, constants, Earth orientation and space weather R10:

Force set	Largest 3D difference
Point mass	0.6 mm
Gravity field, third bodies, radiation pressure, tides, relativity, in-track thrust	Under 1 cm
With drag (NRLMSISE-00 or JB2008, daily space weather, drag rate)	1.6 cm

State-transition matrices agree within 2e-5 and parameter sensitivities within 1.4e-4 (relative). What remains is integration error on both sides, about 1e-11 of the distance flown. Integrating Jacchia's diffusion equations numerically reproduces the closed forms of Jacchia-Roberts R45 to 1.6e-4.

Accuracy against precise orbits

Experiment V1 starts the propagator from ILRS orbits of four geodetic spheres (LAGEOS-1/2 and ETALON-1/2; 16 seeds) and IGS orbits of 32 GPS satellites (96 seeds), adds forces one set at a time, and compares each 72-hour run with the same product V1. GPS radiation pressure is a deliberately crude cannonball. Median (largest) 3D error:

Force set	Spheres, 24 h	Spheres, 72 h	GPS, 24 h	GPS, 72 h
E-a: point mass	46 km (396 km)	110 km (1,213 km)	25 km (40 km)	75 km (170 km)
E-b: zonal harmonics to degree 20	1.3 km (2.2 km)	3.4 km (8.7 km)	2.1 km (6.5 km)	6.0 km (94 km)
E-c: E-b with the Sun and Moon	257 m (2.1 km)	677 m (6.2 km)	331 m (868 m)	983 m (89 km)
E-d: E-c with radiation pressure	246 m (2.1 km)	670 m (6.2 km)	277 m (661 m)	853 m (89 km)
E-e: E-d with the full 20×20 field and IERS Earth orientation	6.2 m (15 m)	18.6 m (51 m)	29.5 m (117 m)	92 m (90 km)
E-f: E-e with IERS 2010 solid tides and relativity	4.8 m (10 m)	13.9 m (31 m)	28.5 m (117 m)	91 m (90 km)

The GPS maxima beyond a day are one satellite, NORAD 35752, in every force set. Two of four pre-registered checks failed: an added force did not reduce every seed's error (14 violations, all without tesseral harmonics), and a 20×20 field without the Sun and Moon beat zonals alone for only 53 of 96 GPS seeds. The two-body return (0.8 mm) and the E-d bound at 24 hours (246 m against 500 m) passed.

HPOP's resident path, which holds the catalog in memory, reproduces its execution path to 0.052 mm over four days, at 0.645 s per object-day on one thread R10.

8 Association and selection

Association asks whether records describe one object; selection picks a solution for an epoch, interval and use; estimation derives a new one. Candidates come from provider crosswalks, international designators and native-identifier histories, Vimpel's datefirst fields included; names and bare numbers are not join keys. Compatibility is tested on normalized ephemerides over shared intervals. Each candidate link is compatible, rejected, ambiguous or insufficient, and accepted links stay reversible. Selection takes, per regime and horizon, the source with the lowest median error on a training window of at least 30 samples E3.

Rule	Measured	Evidence
Selection by evidence	GPS: 3.5 cm at issue (section 5.1)	E3
Fusion of sources	Never better than selection: 1.07–1.17 times its error in GPS at 1–7 days, equal elsewhere	E3
Covariance intersection R46 of one provider's consecutive OMMs	Collapses onto one set (0.90–1.00 of selection's error); naive fusion is overconfident (mean $d^2/3$ 3.3–700)	E2b

9 SP message interchange

The VCM carries an SP state, its force settings and an equinoctial covariance R39. SDN's adapter reads a VCM into an HPOP request and writes the result back as a VCM, transforming the covariance with an exact Jacobian computed by forward-mode dual numbers R47. The format states no covariance units, and no public definition was found. Four messages settle them: the survey's ISS sample and SP messages for a geostationary, a GPS and an eccentric (e = 0.59) orbit.

Covariance rows	Reading	Evidence
Mean motion	Relative, dn/n; covariance scaled by max(1, weighted RMS)²	Every printed U, V and W sigma reproduced within 1 %; no absolute unit does (radians per 1000 s misses two radial sigmas by 1.7 and 2.0 times)
B and AGOM	Fractions of the coefficients (1.5–9.7 %)	The GPS message against ESA's final orbit for one day: radial and in-track RMS 0.46 and 0.64 of their sigmas, cross-track 1.27; read as printed, the sigmas overstate the error 7 to 20 times

A round trip through VCM text in SP's number format recovers the covariance to 2e-5 of its sigmas. A VCM naming JBH09 runs as JB2008 on SET's indices, which the caller supplies.

10 Publication and reproduction

Publication is transactional: acquire and hash the bytes, verify, normalize, validate, sign a manifest, then announce the new catalog head. Consumers read one complete, immutable snapshot; each update names the object, the prior and new solutions, the epoch, the recipe and the provenance. Content addressing and independent replication keep a snapshot retrievable from any node that holds it.

An accepted numerical solution can carry an SGP4 companion: mean elements fitted to it over a declared interval and scored on a held-out stretch before release, as CelesTrak's supplemental GP fits operator ephemerides R48 and the 18th Space Defense Squadron's eGP fits SP states R49. The fit RMS that CelesTrak publishes for each set is the benchmark.

Each experiment's numbers come from a run manifest: the repository and module commits, each module's WebAssembly SHA-256 and the SHA-256 of every input; reports are generated from metrics R1. The

experiments' public site runs the same modules in a web browser: a V1 seed reproduces the committed result exactly, and any Orekit case re-runs from its exact request. Space-Track element sets, SpaceX and Planet files and SET's indices are cited by URL and SHA-256, not copied.

Evidence	Commit
Experiments V1, E1–E8, E13 and E14	orbit-accuracy-experiments `357d992d6c62daac4ccbb2fd230634ac292192b8`
HPOP 1.3.0, its Orekit suite, screen and covariance calibration; maneuver detection 0.3.0	space-data-network-modules `d4a0b1155f64ac088694f9758c32e3b691952674`
HPOP in V1	space-data-network-modules `750b6399173a2678b8e8d0885557976d17a9ad5c`
VCM adapter	space-data-network-modules `1879d8c5c17c061e157c100723a8423c59fd98e2`
Vimpel conversion and refinement	space-data-network-modules `49e159d7003cac3d3e65b03170f11909fa86dd2c`

HPOP's WebAssembly binary in V1, SHA-256:

`473d527175de6606ccf305c45b37218288634fa9de2dcab0d72564a190bdabd1`

11 Measured accuracy

Median 3D position error against precise orbits, by object class and age of the product:

Class (objects)	Product	At issue	1 day	3 days	7 days	Evidence
GPS (32)	Ultra-rapid orbit, propagated	3.5 cm	30 m	104 m	336 m	E3
GPS (32)	Public OMM, SGP4	1.57 km	1.61 km	1.68 km	1.53 km	E3
Laser-ranged LEO (5)	HPOP fitted to laser ranges	7 m	9 m	17 m	42 m	E6
Laser-ranged LEO (5)	Public OMM, SGP4	472 m	485 m	482 m	685 m	E6
Laser-ranged LEO, 800–1,500 km (5)	Public OMM, SGP4	314 m	326 m	326 m	569 m	E3
Laser-ranged MEO (5)	Public OMM, SGP4	411 m	438 m	395 m	413 m	E3
LEO, 400–800 km (6)	Public OMM, SGP4	566 m	597 m	760 m	1.24 km	E3

The HAC's position errors are not published R2. The public figures: element sets in the HAC's own LEO conjunction regime (perigee above 500 km) err by 0.6–3 km from epoch to 72 hours R49; one ASW covariance shown as an illustration has σ of 8.3 m radial, 332 m in-track and 2.1 m cross-track at epoch R37; HASDM's density spread is 0.078 against JB2008's 0.148 (section 5.4).

12 What conjunction assessment consumes

Conjunction screening R50 takes from one catalog snapshot, per object:

Input	From the catalog
Trajectory	The OMM for SGP4, or one propagator's trajectory; one propagator per run
Covariance	Supplied by the operator, fitted, or modeled (section 6), with its calibration label
Hard-body radius	Supplied, half the catalog size, from radar cross-section, or the default, with its basis

Input	From the catalog
Provenance	Snapshot, source records and recipe

An OMM supplies no covariance. A collision probability needs relative-state uncertainty with independent calibration and stated correlation, geometry and hard-body radius; without it the screen reports only the covariance-free maximum probability.

Appendix A. Terms

Term	Meaning
ASO	Anthropogenic space object
OMM, GP	CCSDS Orbit Mean-elements Message: SGP4 mean elements, also distributed in the legacy two-line format; general perturbations element set
SGP4, B*	The analytic propagator of OMMs; its drag term
HPOP	High-precision orbit propagation: numerical integration of the full force model of section 7
Provider claim	An attributed product that already embeds a provider's models and inference
Level 0 to 3	Processing levels of orbit data (section 2)
Ultra-rapid, final orbits	GNSS analysis-centre orbits issued in near real time with a prediction; definitive orbits issued about two weeks later
SLR, ILRS	Satellite laser ranging; the International Laser Ranging Service
TEME, EME2000, GCRF, ITRF	SGP4's output frame; the J2000 mean equator and equinox; the Geocentric Celestial Reference Frame; the terrestrial frame
d^2, mean $d^2/3$	Squared Mahalanobis distance of an error under its covariance; the mean of $d^2/3$ is 1 for a realistic covariance
DCA, DCP	HASDM's Dynamic Calibration of the Atmosphere; the Dynamic Consider Parameter for density-forecast error
HASDM, JB2008, JBH09	High Accuracy Satellite Drag Model; the Jacchia-Bowman 2008 density model; JB2008 with HASDM's calibration
SP, HAC, ASW	Special Perturbations orbit determination and propagation; the High Accuracy Catalog; the Astrodynamics Support Workstation
VCM	SP Vector Covariance Message
Supplemental GP, eGP	CelesTrak's element sets fitted to operator ephemerides; the 18th Space Defense Squadron's element sets fitted to SP states
AOE	SpaceMapper's catalog of element sets from its own sensors
Covariance intersection	A fusion rule that stays consistent under unknown correlation

Data credits

- IGS products, courtesy of the International GNSS Service and its analysis and data centres (Johnston, Riddell and Hausler 2017, doi:10.1007/978-3-319-42928-1). © Navigation Support Office at ESA/ESOC 2026.
- GFZ rapid MGEX orbits of the IGS Multi-GNSS Experiment (Montenbruck et al. 2017, doi:10.1016/j.asr.2017.01.011), obtained from NASA CDDIS (Noll 2010, doi:10.1016/j.asr.2010.01.018).
- ILRS data and products (Pearlman et al. 2019, doi:10.1007/s00190-019-01241-1), from the EUROLAS Data Center, DGFI-TUM (Noll et al. 2019, doi:10.1007/s00190-018-1207-2). NERC Space Geodesy Facility orbits; contains British Geological Survey materials © UKRI 2026, Open Government Licence v3.0.
- Contains modified Copernicus Sentinel data 2026.

- © ESA 2026; data provided by the European Space Agency: SW_SP3xCOM_2_ (doi:10.57780/esa-8579355), SW_DNSxPOD_2_ (doi:10.57780/esa-ef59315), GF_DNSxACC_2_ (doi:10.57780/esa-2559fd1).
- Licata, Mehta, Tobiska, Bowman and Pilinski (2021), CHAMP and GRACE-A density estimates with associated HASDM and JB2008 predictions, Zenodo, doi:10.5281/zenodo.4602380, CC BY 4.0.
- Source: USSPACECOM / 18th Space Defense Squadron, via Space-Track.org.
- SatNOGS Network observations (Libre Space Foundation; the station and observer named in each record), https://network.satnogs.org/, CC BY-SA 4.0.
- SpaceX Starlink public ephemerides, https://api.starlink.com/public-files/ephemerides/ (SpaceX). Planet Labs PBC orbital ephemerides, https://ephemerides.planet-labs.com/, © Planet Labs PBC, CC BY-NC 4.0. ISS trajectory: NASA JSC Flight Operations Directorate (U.S. Government work). China Manned Space Engineering Office, https://www.cmse.gov.cn/. Intelsat (SES) public ephemerides, https://my.intelsat.com/ephemeris/public. EUMETSAT.
- JSC Vimpel, http://spacedata.vimpel.ru/. SpaceMapper, https://spacemapper.cn/. CelesTrak, https://celestrak.org (T. S. Kelso). Data from J. McDowell, planet4589.org.
- The JB2008 solar and geomagnetic indices are provided for scientific use courtesy of Space Environment Technologies (https://spacewx.com/jb2008/). Kp, ap, Ap: GFZ Helmholtz Centre for Geosciences, CC BY 4.0 (Matzka et al. 2021, doi:10.5880/Kp.0001). Sunspot number: WDC-SILSO, Royal Observatory of Belgium, Brussels, CC BY-NC 4.0. F10.7: Dominion Radio Astrophysical Observatory and Natural Resources Canada. NOAA Space Weather Prediction Center (U.S. Government material).
- IERS EOP 20 C04 (IERS Earth Orientation Centre, Observatoire de Paris); finals2000A (IERS Rapid Service/Prediction Centre, U.S. Naval Observatory). JPL planetary ephemeris DE440 (Park et al. 2021, doi:10.3847/1538-3881/abd414), via NASA/JPL NAIF. EGM2008: NGA (Pavlis et al. 2012, doi:10.1029/2011JB008916). EGM96: NASA GSFC and NIMA (Lemoine et al. 1998). FES2004 (Lyard et al. 2006) as distributed with the IERS Conventions (2010).

References

R1

Edgesource. Orbit accuracy experiments: frozen plans, run manifests, generated reports and data licenses, with a browser site that re-runs them. Commit 357d992d6c62daac4ccbb2fd230634ac292192b8. Repository, site

R2

Edgesource. HAC data sources: 97 public and login-only sources, their access, terms and use, and what is public about the HAC. Inventory

R3

Licata, R. J., Mehta, P. M., Tobiska, W. K., Bowman, B. R., and Pilinski, M. D. CHAMP and GRACE-A density estimates with associated HASDM and JB2008 predictions. Zenodo, 2021. doi:10.5281/zenodo.4602380

R4

SpaceMapper. AOE catalog and orbital database. AOE catalog

R5

JSC Vimpel and Keldysh Institute of Applied Mathematics. Orbit parameters of newly detected HEO space debris objects. Bulletin

R6

Vallado, D. A. Fundamentals of Astrodynamics and Applications, 4th edition. Microcosm Press, 2013.

R7

Edgesource. Vimpel epoch conversion, refinement study and 64-object audit. SDN modules repository, commit 49e159d7003cac3d3e65b03170f11909fa86dd2c, `analysis/catalog-composer/docs/`. Private repository, available on request (tj@edgesource.com).

R8

Vallado, D. A., Crawford, P., Hujsak, R., and Kelso, T. S. Revisiting Spacetrack Report #3. AIAA 2006-6753, 2006. Paper

R9

Hoots, F. R., and Roehrich, R. L. Models for Propagation of NORAD Element Sets. Spacetrack Report No. 3, 1980. Report

R10

Edgesource. SDN modules repository (space-data-network-modules), commit d4a0b1155f64ac088694f9758c32e3b691952674: `propagator/hpop/README.md` (force model, Orekit suite, resident path, catalog cost), `analysis/gp-error-model/docs/hpop-calibration-2026-08.md`, `analysis/conjunction-assessment/docs/hpop-full-force-screen-2026-10.md`, `analysis/maneuver-detection/README.md`, `docs/studies/maneuver-detection.md`. Private repository, available on request (tj@edgesource.com).

R11

Petit, G., and Luzum, B., editors. IERS Conventions (2010). IERS Technical Note 36. Technical note

R12

Johnston, G., Riddell, A., and Hausler, G. The International GNSS Service. Springer Handbook of Global Navigation Satellite Systems, 967–982, 2017. Products

R13

Pearlman, M. R., et al. The ILRS: approaching 20 years and planning for the future. Journal of Geodesy 93, 2161–2180, 2019. doi:10.1007/s00190-019-01241-1

R14

Dyreby, K., Caldas, F., and Soares, C. Analyzing Data Quality and Decay in Mega-Constellations: A Physics-Informed Machine Learning Approach. arXiv:2510.11242, 2025. Preprint

R15

Jankovic, D. How long can you trust a Starlink TLE? An empirical comparison of SGP4 and high-fidelity propagation against operator-updated truth across a megaconstellation. arXiv:2605.19850, 2026. Preprint

R16

Tapley, B. D., Schutz, B. E., and Born, G. H. Statistical Orbit Determination. Elsevier Academic Press, 2004.

R17

Ly, D., Lucken, R., and Giolito, D. Correcting TLEs at epoch: application to the GPS constellation. Journal of Space Safety Engineering 7(3), 2020.

R18

Hallgarten La Casta, M. I., and Amato, D. Generalisable bias correction of two-line element sets in the medium Earth orbit regime. 9th European Conference on Space Debris, 2025.

R19

Levit, C., and Marshall, W. Improved orbit predictions using two-line elements. Advances in Space Research 47(7), 1107–1115, 2011. arXiv:1002.2277

R20

Bowman, B. R., Tobiska, W. K., Marcos, F. A., Huang, C. Y., Lin, C. S., and Burke, W. J. A New Empirical Thermospheric Density Model JB2008 Using New Solar and Geomagnetic Indices. AIAA 2008-6438, 2008. doi:10.2514/6.2008-6438

R21

Storz, M. F., Bowman, B. R., Branson, J. I., Casali, S. J., and Tobiska, W. K. High accuracy satellite drag model (HASDM). Advances in Space Research 36(12), 2497–2505, 2005. doi:10.1016/j.asr.2004.02.020

R22

Doornbos, E. Empirical Model Calibration. In Thermospheric Density and Wind Determination from Satellite Dynamics, Springer Theses, 2012. doi:10.1007/978-3-642-25129-0_6

R23

Picone, J. M., Emmert, J. T., and Lean, J. L. Thermospheric densities derived from spacecraft orbits: accurate processing of two-line element sets. Journal of Geophysical Research 110(A3), 2005. doi:10.1029/2004JA010585

R24

Gondelach, D. J., and Linares, R. Real-time thermospheric density estimation via two-line element data assimilation. Space Weather 18(2), e2019SW002356, 2020. doi:10.1029/2019SW002356

R25

Gershman. Near Real-Time Thermospheric Density Monitoring from Public Satellite Tracking and Space Weather Data. NASA, 2026. doi:10.64631/vfnq1575

R26

Ou, Zhong, Hao, Li, et al. Near-Real-Time Global Thermospheric Density Variations Unveiled by Starlink Ephemeris. Remote Sensing 17(9), 1549, 2025. doi:10.3390/rs17091549

R27

Yamamoto. Tomography of thermospheric density from Starlink Ephemeris: initial report. Earth, Planets and Space 78, 2026. doi:10.1186/s40623-026-02509-5

R28

Fitzpatrick, Sutton, Pilinski, and Palo. Harnessing Satellite Constellations as Signals of Opportunity for an Enhanced Specification of the Satellite Drag Environment in the Critical LEO Regime. Space Weather, 2026. doi:10.1029/2025SW004611

R29

Gondelach, D. J., and Linares, R. Real-Time Thermospheric Density Estimation via Radar and GPS Tracking Data Assimilation. Space Weather 19, e2020SW002620, 2021. doi:10.1029/2020SW002620

R30

Lemmens, S., and Krag, H. Two-Line-Elements-Based Maneuver Detection Methods for Satellites in Low Earth Orbit. Journal of Guidance, Control, and Dynamics 37(3), 860–868, 2014. doi:10.2514/1.61300

R31

Willsky, A. S., and Jones, H. L. A generalized likelihood ratio approach to the detection and estimation of jumps in linear systems. IEEE Transactions on Automatic Control 21(1), 108–112, 1976. doi:10.1109/TAC.1976.1101146

R32

Zaidi, W. H., and Hejduk, M. D. Earth Observing System Covariance Realism. AIAA/AAS Astrodynamics Specialist Conference, AIAA 2016-5628, 2016. doi:10.2514/6.2016-5628

R33

Osweiler, V. P. Covariance estimation and autocorrelation of NORAD two-line element sets. MS thesis, Air Force Institute of Technology, AFIT/GSS/ENY/06-M09, 2006. Thesis

R34

Geul, J., Mooij, E., and Noomen, R. TLE uncertainty estimation using robust weighted differencing. Advances in Space Research 59(10), 2522–2535, 2017. doi:10.1016/j.asr.2017.02.038

R35

Thompson, Gossner, Sais, and Cunningham. Fast covariance propagation for two-line element sets. AMOS Technical Conference, 2019. doi:10.64861/twqd5985

R36

Verhaeghe, De Clerk, Lauwens, Delabie, Vandepitte, and Vandenbussche. Predicting TLE covariance using transformer-based time series forecasting. Small Satellite Conference, SSC25-IV-03, 2025. Paper

R37

Hejduk, M. D. ASW Covariance Introduction and Formation. NASA, 2016. NTRS 20160005042

R38

Hejduk, M. D. Three CA-Related Covariance Issues and Their Solutions. NASA, 2019. NTRS 20190026790

R39

Space Data Standards. Legacy message survey: the Vector Covariance Message format and sample; PRW and VCM schemas. Survey

R40

NASA Goddard Space Flight Center. General Mission Analysis Tool (GMAT), atmosphere models. Project

R41

Pachura, D., and Hejduk, M. D. Atmospheric Model at JSpOC. NASA, 2016. NTRS 20160005044

R42

National Research Council. Continuing Kepler's Quest: Assessing Air Force Space Command's Astrodynamics Standards. The National Academies Press, 2012. doi:10.17226/13456

R43

Berry, M. M., and Healy, L. M. Implementation of Gauss–Jackson Integration for Orbit Propagation. Journal of the Astronautical Sciences 52(3), 331–357, 2004. Paper

R44

CS GROUP. Orekit 13.1, open-source space flight dynamics library, Apache License 2.0. Project

R45

Roberts, C. E., Jr. An analytic model for upper atmosphere densities based upon Jacchia's 1970 models. Celestial Mechanics 4, 368–377, 1971. doi:10.1007/BF01231398

R46

Julier, S. J., and Uhlmann, J. K. A non-divergent estimation algorithm in the presence of unknown correlations. Proceedings of the American Control Conference, 2369–2373, 1997.

R47

Edgesource. VCM adapter (`analysis/vcm-adapter`). SDN modules repository, commit 1879d8c5c17c061e157c100723a8423c59fd98e2. Private repository, available on request (tj@edgesource.com). Parity record: vcm-parity.md

R48

CelesTrak. Supplemental GP element sets: methodology for fitting operator ephemerides with SGP4. Supplemental GP

R49

NASA. Spacecraft Conjunction Assessment and Collision Avoidance Best Practices Handbook, Vol. 2, Technical Appendices, August 2026. NTRS 20260000457

R50

Koury, A., and Jah, M. K. Fast All-vs-All Conjunction Screening. Space Data Network technical whitepaper. Paper